Literature Status: A Weak Tile with Neither a Tiling Complement nor a Spectrum

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Status: literature already answered. Open Problem 7.4 in arXiv:2209.04540 was explicitly answered in the affirmative by Kiss–Londner–Matolcsi–Somlai, arXiv:2410.04948. This note records that identification; it claims no new theorem.

1 Original question

Kolountzakis, Lev, and Matolcsi's *Spectral sets and weak tiling*, arXiv:2209.04540, asks the following in Section 7.4 on arXiv PDF page 18 [1]. Here "weakly tiles its complement" means that there is a positive locally finite measure ν with $\mathbf{1}_\Omega * \nu = \mathbf{1}_{\mathbb{R}^d \setminus \Omega}$ almost everywhere.

Source Open Problem 7.4. *Does there exist a bounded measurable set $\Omega \subset \mathbb{R}^d$ which can weakly tile its complement by translations, but which can neither tile the space properly by translations nor be a spectral set?*

This is the fourth and last open problem in the source paper.

2 Explicit separate answer

The abstract and introduction of Kiss, Londner, Matolcsi, and Somlai's separate paper *A lonely weak tile*, arXiv:2410.04948, explicitly say that they answer the question raised in the source paper, which they cite as reference [19] [2]. On arXiv PDF page 2 they state the target question and announce an affirmative answer.

Separate later answer. *There is a finite union $C(k)$ of unit cubes in $\mathbb{R}^5$ which weakly tiles its complement, but is neither a proper translational tile nor a spectral set.*

Section 3 first constructs a finite-group set P_t which weakly pd-tiles the ambient group but is neither spectral nor a tile. In the final paragraph on arXiv PDF page 7, the authors transfer the construction to $C(k) = P(k) + [0, 1)^5 \subset \mathbb{R}^5$. For sufficiently large k, standard lifting results preserve non-tiling and non-spectrality, while the lifted pd-tiling measure has a unit mass at the origin; removing that mass gives the required weak tiling of $C(k)^c$.

The provenance is explicit rather than inferred: the later paper names and cites the source question, and Máté Matolcsi coauthored both papers.

3 Scope and search status

This completely answers Open Problem 7.4 in the affirmative. It does not answer the other three questions in Section 7: nowhere-dense spectral sets in dimensions $d \geq 2$, spectrality of the second factor in certain product sets, or whether every extremal weak-tiling measure for a convex body comes from a proper tiling.

A bounded search through 9 August 2026 checked the run indexes, the exact wording "weakly tiles its complement", and arXiv/literature searches for weak tiles that are neither spectral nor proper tiles. The decisive exact match was arXiv:2410.04948. No new mathematical result is claimed here.

References

[1] M. N. Kolountzakis, N. Lev, and M. Matolcsi, *Spectral sets and weak tiling*, Sampl. Theory Signal Process. Data Anal. **21** (2023), article 31; arXiv:2209.04540. doi:10.1007/s43670-023-00066-w.

[2] G. Kiss, I. Londner, M. Matolcsi, and G. Somlai, *A lonely weak tile*, Expositiones Math. **44**(1) (2026), article 125636; arXiv:2410.04948. doi:10.1016/j.exmath.2024.125636.